

# Multilingual Fake News Detection Using XLM-RoBERTa: Zero-Shot, Few-Shot, and Fine-Tuning Comparison

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## Abstract

I investigate automatic fake news detection in Turkish news headlines by comparing three learning regimes built on XLM-RoBERTa: zero-shot NLI inference, 32-shot fine-tuning, and full supervised fine-tuning on a combined 8,078-sample Turkish corpus. Full fine-tuning achieves 93.1% accuracy and 0.969 AUC-ROC; zero-shot reaches 65.8% with no labeled data. A cross-lingual qualitative probe reveals interpretable heuristics the model learned that partially generalize to English.

## 1. Problem Definition, Motivation, and Dataset

Fake news detection in Turkish is an underserved problem despite active misinformation on Turkish social media. I take a **pattern-based** approach: the model learns linguistic and stylistic signals associated with fake or satirical content rather than fact-checking against a knowledge base. The task is binary classification (*fake* vs. *real*) on headline text only, forming the LLM component of a larger multimodal detection project.

I combined three Turkish-language sources (Table 1). Real news comes from Hürriyet, Milliyet, CNN Türk, and TRT; ground-truth labels from Teyit.org and the known satirical nature of Zaytung. The corpus was split 80/20 stratified by label (6,462 train / 1,616 test). Zero-shot evaluation used a 500-sample subset due to inference speed.

| Dataset                               | Samples      | Fake / Real          |
|---------------------------------------|--------------|----------------------|
| TR-FakeNews (headlines, fact-checked) | 5,326        | 2,845 / 2,481        |
| MiDe22 (social media posts)           | 2,401        | 1,732 / 669          |
| Zaytung (satirical news, scraped)     | 351          | 237 / 114            |
| <b>Total</b>                          | <b>8,078</b> | <b>4,814 / 3,264</b> |

Table 1: Combined training corpus.

## 2. LLM Methodology

All regimes use XLM-RoBERTa [1], a multilingual transformer pre-trained on 100 languages, making it suitable for Turkish without a language-specific model.

**Zero-shot (NLI).** I use `joeddav/xlm-roberta-large-xnli` [2] with no task-specific training. Each headline is the NLI premise; I score two Turkish hypotheses via the template “*Bu haber {gerçek/sahte}.*” (“This news is {real/fake}.”) and threshold the real-entailment score at 0.5.

**Few-shot (32-shot).** I add a 2-class head to `xlm-roberta-base` and fine-tune on 64 labeled examples (32 per class) for 10 epochs, batch size 8,  $\text{lr} = 2 \times 10^{-5}$ . This simulates low-annotation deployment.

**Full fine-tuning.** Same architecture, trained on all 6,462 examples for up to 4 epochs (batch 16,  $\text{lr} = 2 \times 10^{-5}$ , 100 warmup steps, `EarlyStoppingCallback` patience 2 on macro F1). The best checkpoint was at **epoch 1**, indicating rapid convergence from the pre-trained representations.

## 3. Experiments and Results

### 3.1. Quantitative Comparison

Figure 1 shows all metrics side by side. Full fine-tuning (93.1% accuracy, AUC 0.969) far outperforms the data-scarce regimes; the ROC curves make the separation especially clear. Notably, even 64 labeled examples yield 85.5% AUC-ROC in the few-shot setting, showing that XLM-RoBERTa’s multilingual representations already encode strong credibility priors.

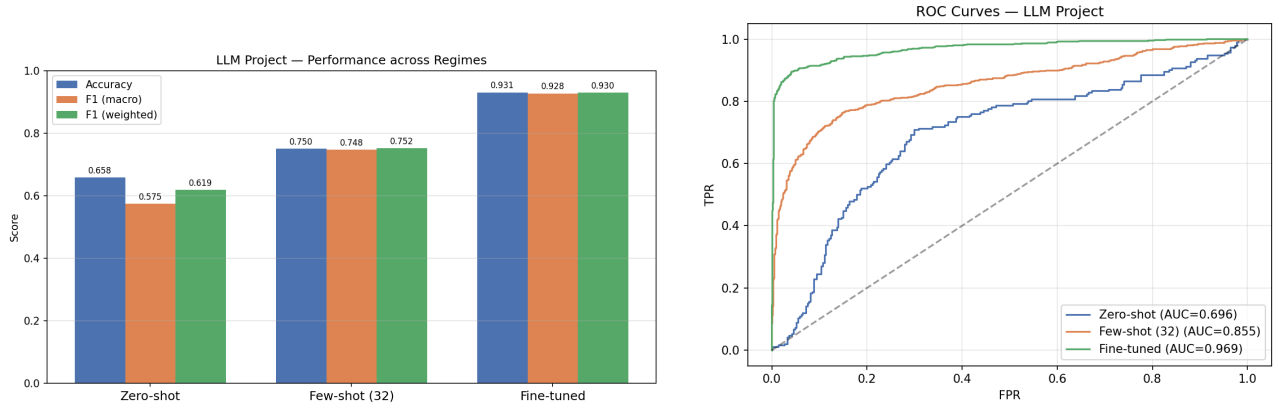


Figure 1: Left: accuracy, F1-macro, and F1-weighted across regimes. Right: ROC curves (AUC: zero-shot 0.696, few-shot 0.855, fine-tuned 0.969).

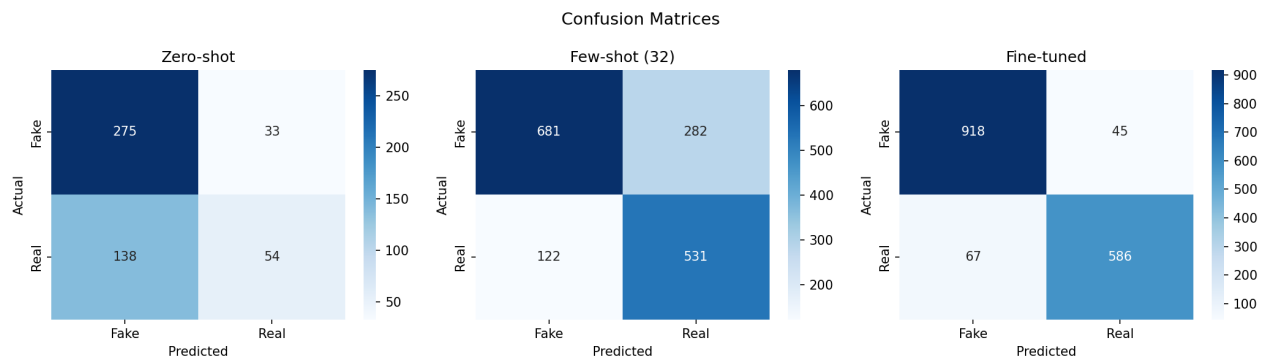


Figure 2: Confusion matrices for all three regimes. Zero-shot is heavily biased toward predicting fake (138 real headlines mislabeled); fine-tuning reduces this to 67.

### 3.2. Error Analysis via Confusion Matrices

The zero-shot model predicts almost everything as fake: only 54 of 192 real samples are correctly identified. The few-shot regime partially corrects real-class recall (531/653 correct) but still misses 282 fake samples. Full fine-tuning achieves 918/963 fake and 586/653 real correctly, with errors roughly balanced between classes.

### 3.3. Cross-Lingual Qualitative Evaluation

I applied the fine-tuned model to 20 hand-crafted headlines (5 TR real, 5 TR fake, 5 EN real, 5 EN fake), achieving 14/20 overall (TR: 9/10, EN: 5/10). Three error patterns emerged:

- Extreme numbers trigger false positives.** “Merkez Bankası politika faizini yüzde 45’te sabit tuttu” (CB holds rate at 45%) was flagged FAKE with 73% confidence — the model associates large percentages with satire.
- Authoritative framing suppresses detection.** Headlines starting with “Uzmanlar:” or “Hükümet duyurdu:” followed by absurd claims were both misclassified as REAL — institutional framing overrides content plausibility.
- Absurdity generalizes cross-lingually; credibility does not.** All 5 English fake headlines (absurdist) were correctly classified, while all 5 English real headlines were misclassified as fake — real-news style is out-of-distribution for a Turkish-trained model.

## 4. Discussion, Limitations, and Conclusion

The zero-shot NLI baseline underperforms because the Turkish hypothesis framing does not capture satirical style signals; a more descriptive prompt would likely help. The 32-shot result (85.5% AUC-ROC with 64 examples) is the most practically interesting finding: it suggests that for annotation-constrained

settings, XLM-RoBERTa needs only a handful of labels to become useful.

**Limitations:** The dataset is headline-only (short text); performance on full articles may differ. Class imbalance ( $\approx 60\%$  fake) means macro F1 and AUC are more reliable than accuracy. The model is Turkish-centric and cannot verify factual accuracy, so a well-disguised fake using formal language can still fool it.

In summary, full fine-tuning on 8,078 Turkish samples yields 93.1% accuracy and 96.9% AUC-ROC, and even a 32-shot regime is viable for low-resource deployment. Qualitative analysis reveals heuristics — numerical extremity, institutional framing, absurdist language — that partially transfer across languages, confirming that multilingual pre-training is a strong foundation for Turkish misinformation detection.

## References

- [1] Conneau, A. et al. (2020). Unsupervised cross-lingual representation learning at scale. *ACL 2020*, pp. 8440–8451.
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- [3] Altunbey Özbay, F., & Birturk, A. (2020). Turkish fake news dataset. *Mendeley Data*.
- [4] Kula, M., Sert, M., & Özçelik, R. (2022). MiDe22: Misinformation detection on Turkish social media. *LREC 2022*.